

Markscheme

Mathematics: analysis and approaches
Practice question bank

60. (a)

A sequence satisfies $u_1 = 4$ and $u_{n+1} = 3u_n - 5$ for $n \geq 1$.

Find the value of u_4 .

$$u_2 = 3(4) - 5 = 7 \quad (M1)$$

$$u_3 = 3(7) - 5 = 16 \quad \mathbf{A1}$$

$$u_4 = 3(16) - 5 = 43 \quad \mathbf{A1}$$

[3 marks]

(b)

Hence find the value of $\sum_{r=1}^4 u_r$.

$$4 + 7 + 16 + 43 \quad (M1)$$

$$= 70 \quad \mathbf{A1}$$

[2 marks]